Gennadii Filatov
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Education

ITMO University (St. Petersburg, Russia) **2025-present**
M.Sc. in Applied Mathematics and Computer Science with a specialization in Big Data and Machine Learning ***(GPA 4.00/4.00)***

Peter the Great Saint Petersburg Polytechnic University (St. Petersburg, Russia) **2021-2025**
B.Sc. in Physics with a specialization in physics of elementary particles and the atomic nucleus ***(GPA 3.92/4.00)***
Thesis: "Precision measurement of cathode wire positions in the high-pressure proportional chamber system for the experiment determining the proton charge radius via elastic electron-proton scattering" (Advisor: Dr. Polina Kravchenko)

Petersburg Nuclear Physics Institute (Gatchina, Russia) **Mar 2025**
57th Winter School on Nuclear and Particle Physics

Peter the Great Saint Petersburg Polytechnic University (St. Petersburg, Russia) **Sep-Dec 2023**
Minor course in Mathematical Physics by prof. Irina Suslova (passed the final exam with an excellent mark)

Peter the Great Saint Petersburg Polytechnic University (St. Petersburg, Russia) **Sep 2022- Nov 2023**
Program of additional education – Developer of the Python application software (passed the final exam with an excellent mark)

Work Experience

Senior laboratory assistant **Jul 2023-present**
Led research optimizing PRES experiment data analysis: developed innovative methods to process wire chamber images and mitigate systematic errors, achieving 30% higher precision in proton charge radius measurement. PRES experiment.
Department: High energy physics department
Contact: Dr. Polina Kravchenko

Awards

Awarded Sber Competitive IT Scholarship (2025) | Charitable Foundation "Investment in the Future"
Winner, SPbPU Research Portfolio Competition (2025)
Awarded Enhanced State Academic Scholarship (2022-2025) | Ranked within top 5% of SPbPU students
Awarded "Student of the Year" (2024) | Peter the Great St. Petersburg Polytechnic University (SPbPU)
Awarded IAPS Worldwide Grant (2024) | For participation in the International Conference of Physics Students (ICPS)
Recipient, Polytech Towers Award: Best Project in Adaptive Scientific Activity (2023) | SPbPU
Secured "Polytech Project" Grant (2024) | To develop "SPARK": a platform for soft-skills development among young scientists | SPbPU

Research Interest

- Data-Centric AI & Synthetic Data: Quality assessment, generation, and validation of tabular synthetic data for scientific applications.
- Machine Learning for Science: Development and reliability estimation of ML models for physics, plasma diagnostics, and experimental data analysis.
- Big Data Pipelines in Megascience: End-to-end data processing, statistical analysis, and automation for large-scale experimental facilities (e.g., NICA, LHC).

Research Experience

Data science researcher **Oct 2025-present**
Engineered ridge regression-based calibration method for synthetic tabular data validation, achieving 60-79% model rank preservation across 5 UCI datasets with 44 ML architectures, improving Spearman correlation from 0.39 to 0.97 (CTGAN, Obesity dataset)
Department: Natural Systems Simulations (NSS) laboratory
Contact: Dr. Irina Deeva

Senior laboratory assistant **Jul 2023-Aug 2025**

Led research optimizing PRES experiment data analysis: developed innovative methods to process wire chamber images and mitigate systematic errors, achieving 30% higher precision in proton charge radius measurement. PRES experiment.

Department: High energy physics department

Contact: Dr. Polina Kravchenko

Data science researcher **Feb-Jun 2023**

Engineered a 1D convolutional neural network (Python, TensorFlow/Keras) for high-accuracy filament detection in fusion plasma data; Improved processing speed and accuracy gains over manual methods through automated signal analysis and robust preprocessing.

Department: Laboratory of "Diagnostics of high-temperature plasma"

Contact: Dr. Alexander Yashin

Volunteering & leadership

International Association of Physics Students (St. Petersburg, Russia) **Apr 2023-Apr 2025**

President of IAPS Local Committee in St. Petersburg

Created and managed a strong team of students, which created Event projects and media products for popularization of science in the majority. Coordinator of International education and exchange programs. Have written 3 grants for creating new education projects for physics students in St. Petersburg in the total amount of 17 000 euro. Created educational courses and events that more than 1,500 Polytechnic and St. Petersburg students went through during their 2 years of work, which allowed them to develop soft and hard skills for scientific activities.

Summer scientific school for schoolchildren "A5 school" (St. Petersburg, Russia) **Aug 2022-Jul 2024**

Founder and Head of Education department

Created a team and engaged in recruiting teachers for a summer school to immerse students in the scientific and technical environment. He was engaged in the formation of the educational program and the training of teachers for the course. Editing courses written by teachers. More than 115 St. Petersburg schoolchildren have completed the courses I have created and enrolled in top Russian universities.

The Three Sciences Tournament (St. Petersburg, Russia) **Apr 2024-May 2025**

Head of the Organizing Committee in the North-Western Federal District of Russian Federation

Formed a team of organizers to create tournament assignments, was engaged in conducting and preparing a platform for solving cases in physics, biology and chemistry. The tournament was attended by 12 teams with a total of 60 people. During the tournament, we managed to create a strong community of like-minded people for further development in science and solving joint cases from industrial partners.

Gazprom Industry Olympiad for Schoolchildren (St. Petersburg, Russia) **Oct 2023-Feb 2024**

Member of the Organizing Committee of the Physics Olympiad

I was engaged in the creation of Olympiad problems in physics for schoolchildren in grades 8-11 with an applied bias for the preliminary and final stages of the Olympiad. Subsequently, I checked tasks from different regions.

Polytechnic Community of Physics Students (St. Petersburg, Russia) **Mar 2022-Apr 2025**

I was a part of the student community. He created 4 educational projects for Polytechnic students: a Physics Club, an English-Speaking Club, a Young Scientist's School, and an educational seminar on experimental data processing and statistics. Over the course of 4 years, I have assembled a team of 88 people who continue to promote these projects. In 4 years, he conducted 113 events, through which more than 1,650 students passed.

Presentations/Conferences

Poster presentation, *Precision measurement of the position of cathode wires in a system of proportional high-pressure chamber in an experiment to measure the charge radius of a proton in elastic electron-proton scattering*. (2024) **38th International Conference of Physics Students (ICPS 2024), Georgia**

Presentation, *Status of PRES experiment and precision Cathode Wire Position Measurement in a High-Pressure Proportional Chamber for Proton Charge Radius Experiment*. (2024) **XI All-Russian Youth Scientific Forum "Open Science" with international participation**

Presentation, *Development of a diagnostic method for multi-wire proportional high-pressure chambers created for the PRES experiment*. (2025) **All-Russian Conference on elementary particle and nuclear physics section "Week of Science at the Institute of Physics and Mechanics"**